Sahaj Bhadja

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Summary

R&D-focused software engineer with 2+ years building **C++/Python solutions**, **embedded-style validation workflows**, and **regression test automation** for high-reliability systems and data-intensive products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Designed and implemented **Python services and data interfaces** for 500+ users, reducing query latency from **1.2s to 350ms** through analytical debugging and software optimization.
- Built **testable orchestration workflows** for 5K+ monthly jobs with validation safeguards and rollback controls, improving reliability and release confidence under tight schedules.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **full-stack software features** handling 10K+ daily requests, translating product requirements into maintainable modules with structured validation and documentation.
- Improved **CI/CD and regression coverage** with automated checks and defect triage, reducing release turnaround by **60%** while maintaining production stability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **reproducible benchmarking pipelines** over 100K+ trajectories, enabling systematic test execution and quantitative analysis for complex system behavior research.
- Produced **technical diagnostics and reports** to support design decisions and debugging handoffs between researchers and software engineering collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **backend integration services** with Python/Java, AWS Glue, and S3, improving response time by **30%** and strengthening distributed software reliability.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across Linux/Windows environments, improving pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving response consistency and user-facing system performance.
- Automated **Docker/Jenkins deployment and smoke tests**, reducing deployment time by **50%** and standardizing release quality across environments.

Skills

Languages: **C++**, **Python**, Java, JavaScript/TypeScript, SQL, Bash
Engineering: object-oriented design, requirements analysis, design/code reviews, root-cause debugging
Validation: regression testing, feature validation, test planning, performance and integration checks
Systems: embedded programming concepts, Linux/UNIX, Windows tooling, multithreading fundamentals
Tooling: Git/Team Foundation concepts, Jenkins, Docker, CI/CD, documentation, Agile collaboration

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLFQ algorithms** with runtime metrics and visualization tooling, reducing debugging time by **20%** through deterministic test scenarios.

Instrument Telemetry Validation Harness | *C++, Python, Linux, JSON*

- Built a **prototype validation harness** for test-measurement style telemetry streams with replay and assertions, improving defect isolation and repeatability across firmware/software interfaces.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized REST pipeline** with automated CI/CD validation, improving operational reliability and sustaining production-grade data processing workloads.